

Response to the Editor

Manuscript: *Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation*

Journal: Information Processing & Management

Dear Dr. Jansen,

Thank you for reading the manuscript and for the detailed instructions. We have revised the manuscript and the supplementary material accordingly. Below, each point of your message is quoted in short form, followed by what we did and where. Section numbers refer to the revised manuscript; “Supplement Sn” refers to the revised Supplementary Material.

Summary of the main changes

- **Length.** The manuscript is shortened from about 13,000 to 9,300 words of text, and from 79 to 53 pages in the review layout (34 pages of text, followed by references, declarations and the float pages). The proofs, the experimental details and the secondary results (sparse logs, exposure concentration, stakeholders and regulation) moved to the Supplementary Material.
- **New Section 2** “Research objectives and contributions” with four objectives and four research questions.
- **New Section 7.1** “Datasets” with a table describing all four datasets, and Section 7.2 “Baselines”.
- **New Section 9** “Discussion and implications” with separate theoretical implications, practical implications and limitations.
- **Abstract** rewritten: 216 words, with dataset sizes and numerical results.
- **Highlights** rewritten: five items of 19 to 24 words.
- **Title** changed, to name the field (causal recommendation) and the contribution.

Point-by-point response

1. **Reviewing for IP&M.** Noted. The author will accept review invitations from the journal.
2. **“Highlights are not highlights ... or are too long; about 15 to 30 words each.”** We rewrote the five highlights so that each states one finding or contribution in 19 to 24 words, for example the exact bias of debiased graph propagation beyond one hop, the 18–29% overestimate of utility that the correction removes on a log with known exposure, and the unchanged accuracy over four datasets and up to 26 baselines (file **highlights**). They include the negative result that the guarantee fails when degree weights come from the log. (Elsevier’s general guidance suggests at most 85 characters per highlight; we followed your length instruction.)

3. “The abstract does not highlight the specifics . . . too much background; add details; shorter is better.” The abstract is rewritten from 250 to 216 words. Background is reduced to two sentences. It now gives the data (four datasets, 290 to 15,400 users, up to 10.3 million logged interactions; a semi-synthetic KuaiRec log), the comparison (up to 26 tuned baselines) and the main numbers: the three-hop error reduced by a factor of 4 to 12, the 18–29% utility overestimate removed under oracle conditions and 100–540% overestimate with estimated propensities, accuracy differences of at most 0.0014 nDCG, a loss of up to 0.06 nDCG for the sample-split variant, and the 0.004 nDCG margin on KuaiRec that lies within the seed spread of the best trained model.

4. “An explicit research objective(s), preferably as a separate section.” New **Section 2, Research objectives and contributions**. It states four objectives (O1 estimator, O2 conditions, O3 value, O4 invariance) with four research questions (RQ1–RQ4), gives short answers, and lists the contributions. The Results (Section 8) and the Discussion (Section 9) are organised by the same research questions.

5. “Questions about your dataset/sample; explicitly describe the dataset(s) in the manuscript.” New **Section 7.1, Datasets**, with Table 2 (users, items, size and density of the biased log, what the observation indicator means, the unbiased test set, candidates per user and number of evaluated users) and a paragraph per dataset: Coat (290 users, 300 items, 6,960 ratings; 16 random ratings per user for test), Yahoo! R3 (15,400 users, 1,000 songs, 311,704 ratings; 54,000 random ratings of 5,400 users), KuaiRand-Pure (10,000 sampled users, 7,583 videos, 511,863 impressions; random impressions as test) and KuaiRec (7,176 users, 10,728 items, 10.3 million views; a fully observed $1,411 \times 3,327$ matrix as test). We state how positives are defined, how candidates are chosen, and the metric values of a random ranking. We also state that on Coat and Yahoo! R3 the “exposure” is user self-selection, not a platform’s display policy.

6. “No comparison to a state-of-the-art baseline; baselines stale (use or address not using LLMs); unclear whether baselines were run or taken from the literature; same datasets.” New **Section 7.2, Baselines**. (a) *We ran every baseline ourselves*, on the same four datasets, the same splits and the same validation data; no number is taken from the literature. (b) The comparison includes recent graph, contrastive and debiased models: BSPM (2023), SimGCL, r-AdjNorm and NAVIP (2022), MRDR (2021) and GF-CF (2021), together with LightGCN, EASE, iALS, DR-JL, MACR and PDA; each is tuned per split, and the number of configurations is reported (Supplement S2). (c) A new paragraph explains the baselines we did not run (StableDR, DICE, CausE, AutoDebias) and why. (d) *Large language models*: we did not compare with LLM-based recommenders. Our pipeline uses only interaction data, none of the four benchmarks provides item text together with a random-exposure test set, and the question studied is the estimation of a propagation operator under exposure bias, for which the relevant comparators are graph operators. We state this explicitly in Section 7.2 and as a limitation in Section 9.3. We recognise that this is a genuine limitation and do not claim the method outperforms LLM-based recommenders. (e) We additionally report new controls for the claim that gains come from the DR graph: propagation over the imputation alone, equal-budget comparisons and seed variation (Sections 8.3, 9.3).

7. “Highlight the theoretical and practical implications; how does this research differ from existing work; explicit discussion section.” New **Section 9, Discussion and implications**, with Section 9.1 theoretical implications (propagation-level debiasing must consider the

operator; a guarantee for fixed nuisances is a statement about the whole pipeline, including degree weights; unbiasedness for the full-exposure target does not imply better rankings), Section 9.2 practical implications (when the correction is and is not worth using, what needs true propensities and nuisances independent of the log, what the audit can and cannot show), and Section 9.3 limitations. The difference from existing work is stated in Section 3 (“Debiasing inside graph propagation” and “Unbiased estimation of polynomials under missingness”) and in Section 2: the reduction itself is classical; what is new is its exact form for inverse-propensity and DR graph propagation, the exact bias of the uncorrected operators, and the finding that the guarantees fail under data-dependent degree weights.

8. “Thorough copy editing; many readers are not native English speakers.” We rewrote the introduction, related work, experimental design, results, discussion and conclusion in plain language with shorter sentences, defined the central symbols in the introduction, and edited the prose of the method section for clarity without changing any mathematics, citation, number or result (verified by an automated check of formulas, citations, labels and numbers before and after). We would welcome a further language check at the proof stage.

9. “The manuscript is WAY too long; shorten it.” See the summary: 13,000 $\rightarrow$ 9,300 words of text and 79 $\rightarrow$ 53 pages in the review layout, with 34 pages of text. Concretely: the related-work section is about 40% shorter (1,320 to 770 words); the simulation, accuracy and audit sections are condensed to the findings needed for the four research questions; the variance and clipping theorem with its proof, the experimental details and the proofs of all theorems moved to the Supplementary Material (S1–S3); and the sparse-log experiment, the exposure-concentration and clip analyses, and the stakeholder and regulation discussion moved to Supplement S5.

10. “The reference list needs tidying; missing items or formatting issues; use a standard style (APA).” We audited every cited entry for author, title, venue, year, volume, pages and DOI, completed the missing pages, fixed capitalisation of titles (e.g. Möbius, “A worrying analysis”), and the printed list contains only the 65 works cited in the manuscript. The list now follows APA 7, the style the journal’s reference template is based on: author–year citations, DOIs as links, preprints marked as such, and dataset papers marked “[Data set]”. Two conference papers (ICLR 2023, NeurIPS 2019) have no page numbers.

11. “Review recent IP&M articles on similar topics.” The related work already cites recent IP&M articles on debiasing, fairness and privacy (for example on popularity debiasing from user and item perspectives, controllable popularity bias, invariant debiasing with biased imputation, causal deconfounding of multi-criteria ratings, fairness measurement, provider fairness and differential privacy for matrix factorisation), and we kept them. We restructured the paper along the usual research-article layout (introduction, objectives, related work, method, experimental design, results, discussion and implications, conclusion) and moved supporting material to the supplement.

12. Author guidelines. The abstract is under 250 words, the highlights follow your length instruction, and the title page contains the declarations of funding, competing interests, generative AI use, CRediT roles, ethics and data availability, with the public repository for code and result files.

Other changes made during the revision

- New tests: equivalence (TOST) for the key comparisons, a Nadeau–Bengio test over splits adjusted over the same family as the user-level tests, and a note that identical rankings (Yahoo! R3, KuaiRand) cannot be tested (Supplement S5, Table S19).
- A simulation control that fixes the degree weights a priori shows that cross-fitting alone preserves the correction and that data-dependent degree weights break it (Section 8.1, Table S18).
- Table 1 now separates DRUP-split (assumptions hold) from cross-fitted DRUP (as run, no guarantee).
- The claim that the correction “changes no ranking” is replaced by “no detectable change in accuracy”, and the exposure-invariance audit is described as having limited power.

Points we did not address, and why

- **LLM-based and newer baselines.** Not run (Section 7.2). A comparison needs a dataset with item text and a random-exposure test set.
- **Main-text length.** At 34 pages in the review layout the text is still long for some readers. Further shortening would require removing results that answer the research questions.

We hope the revised manuscript addresses your concerns, and we are grateful for the opportunity to resubmit.

Sincerely,
Quang-Vinh Dang
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